

Unit 6: Unequal Sized Clusters

PPS

- A. Controlling sample size
- B. PPS
- C. Systematic PPS

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A. Controlling sample size

- Potentially large variation in sample size
 - A ratio mean for design-consistent estimation
- More control over sample size. Why?
 - Bias of r
 - Variance of r
- Equalize workloads controlling survey costs

Example: Two-stage selection

- A sample of $n = 36$ employees from $N = 2,160$
 - $f = 36/2160 = 1/60$

Unit	No.	Unit	No.
1	443	6	291
2	162	7	64
3	127	8	70
4	554	9	232
5	115	10	102

Sample size control

- Redefine clusters
- Stratification by size, varying sampling rates by size of cluster stratum
- Give up *epsem* for equal size subsamples
- Probabilities proportionate to size (PPS)

Redefine clusters

- Create 10 clusters of approximately $N/10 = 216$ employees each
- Variation above and below desired cluster size
- Once clusters created, select the sample with uniform sampling fractions

$$f = f_a f_b = \left(\frac{a}{A}\right)\left(\frac{b}{B}\right)$$

$$fN = \left(\frac{1}{60}\right)(2160) = 36$$

Stratify by size

Stratum I		Stratum II	
Unit	No.	Unit	No.
2	162	1	443
3	127	4	554
5	115	6	291
7	64	9	<u>232</u>
8	70		1520
10	<u>102</u>		
	640		

Varying sampling rates

- Obtain *epsem* by choice of first and second stage rates:
- Use in stratum I: $f_{Ia} = \frac{1}{6}$ $f_{Ib} = \frac{1}{10}$
- Use in stratum II: $f_{IIa} = \frac{1}{4}$ $f_{IIb} = \frac{1}{15}$
- Sample size in stratum I ranges from 6 to 17, stratum II from 15 to 37
- $cv(x)$ has been reduced within strata.

Equal subsample size

- Select clusters with equal rates but take equal size subsamples
- Non-*epsem* design

$$f = f_a f_{b|a} = \left(\frac{a}{A}\right)\left(\frac{b}{B_\alpha}\right) = \left(\frac{2}{10}\right)\left(\frac{18}{B_\alpha}\right) = \frac{16}{10 \times B_\alpha} = f_\alpha$$

This should be 36, not 16.

- Knowing B_α for each selected cluster, create weights $w_{\alpha\beta} = f_\alpha^{-1}$
- Constant overall workloads at the price of weighting and increases in sampling variances relative to *epsem* designs

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PPS, or how to achieve *epsem* and equal subsample sizes at the same time

- Goal: Select a two stage sample such that there are equal subsample sizes in each cluster with *epsem*
- We want an overall *epsem* rate $f = n/N$ such that the sampling rate in the second stage is b/B_α

$$f = f_\alpha f_{\beta|\alpha} = \left(f_\alpha \right) \left(\frac{b}{B_\alpha} \right) = \left(\frac{n}{N} \right)$$

- Solving $f_\alpha = \frac{n}{N} \times \frac{B_\alpha}{b} = \frac{ab}{\sum_{\alpha=1}^A B_\alpha} \times \frac{B_\alpha}{b} = \frac{aB_\alpha}{\sum_{\alpha=1}^A B_\alpha}$

PPS implications

- Select clusters with probabilities proportionate to their size, B_α
 - Overall the selection becomes $f = \frac{aB_\alpha}{\sum_{\alpha=1}^A B_\alpha} \times \frac{b}{B_\alpha}$
- Selection probability varies at both first & second stages
 - Overall, *epsem*
 - Large clusters selected with **high probabilities**, *subsampling* at **low rate**
 - Small clusters selected with **low probabilities**, *subsampling* at **high rate**

Selecting clusters with PPS

- Two approaches
 - Simple replicated subsampling – at random and with replacement
 - Systematic sample and without replacement
- For both approaches, need to cumulate (sequentially add up) the size of the clusters

Selecting clusters with PPS

Unit	B_{α}	Cum. B_{α}	Unit	B_{α}	Cum. B_{α}
1	443	443	6	291	1692
2	162	605	7	64	1756
3	127	732	8	70	1826
4	554	1286	9	232	2058
5	115	1401	10	102	2160

Using cumulative counts

- Simple replicated subsampling for $a=2$:
 - Select RN from 1 to 2160 with replacement
 - Find the cluster (Unit) with cumulative sum first exceeding RN
 - Select
 - Repeat
 - Subsampling rate b / B_α requires exactly b
- If cluster selected k times, take $k \times b$ elements
- Subsampling: SRS, systematic, stratified element, even cluster sampling

Calculating subsampling rates

- For example, $RN_1=702$ and $RN_2 = 1744$ select Units 3 and 7
- For Unit 3, $\frac{aB_\alpha}{N} = \frac{2 \times 127}{2160} = \frac{1}{8.504}$
- Yields within $\frac{b}{B_\alpha} = \frac{18}{127} = \frac{1}{7.056}$

Variance estimation

- For PPS, $n=ab$ is fixed. Hence,

$$\bar{y} = \frac{\sum_{\alpha=1}^a \sum_{\beta=1}^b y_{\alpha\beta}}{ab}$$

$$\text{var}(\bar{y}) = \frac{(1-f)}{a} s_a^2$$

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C. Systematic PPS

- Compute zone size: $N/a = 2160/2 = 1080$
 - Select one subsample from each zone size 1080
 - Zone boundary falls 348 "elements" (measure units) through Unit 4; falls in 2 zones
- Select Units by selecting RS from 1 to 1080
 - Identify selection from cumulative counts
 - Add the interval $k = 1080$ to the RS
 - Identify next selected cluster
 - Select one subsample from each at the rate b/B_α

Selecting clusters with PPS

Unit	B_{α}	Cum. B_{α}	Unit	B_{α}	Cum. B_{α}
1	443	443	6	291	1692
2	162	605	7	64	1756
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Within sampling rates

- For example, suppose $RN = 804$
 - Select $\alpha = 4$
 - $RN + I = 804 + 1080 = 1884$ selects $\alpha = 9$
- For $\alpha = 4$, $\frac{aB_\alpha}{N} = \frac{B_\alpha}{N/a} = \frac{554}{2160/2} = \frac{554}{1080} = \frac{1}{1.950}$
- Subsample $\frac{b}{B_4} = \frac{18}{554} = \frac{1}{30.78}$

PPS without replacement

- Large literature on without replacement PPS selection procedures
- Alternatives lead to variance estimation expressions that require knowledge about the joint probabilities of selection
- Horvitz-Thompson estimator and the Yates-Grundy estimator of variance
- Unfortunately, the Yates-Grundy estimator can be negative
- Literature proposes wide variety of designs to overcome this problem

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- B. Stratification
- C. Oversize units
- D. Undersize units

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A. Estimated size measures

- Suppose that the exact count of elements -- B_α -- in each cluster is unknown
- But measure that approximates the number of units (that is, the size) in each cluster -- Measure of Size -- Mos_α
- Example: Do not know the current exact count of employees for each Unit
 - Do know the number counted for each Unit at the last payroll one month ago

Example

Unit	Last payroll	Now
1	443	460
2	162	172
3	127	130
4	554	554
5	115	125
6	291	310
7	64	68
8	70	74
9	232	246
10	102	141
Total	2160	2280

- Probabilities proportionate to estimated size: *epsem* design in two stages
- Over-sample large clusters relative to small
- Select clusters with the rate $\frac{aMos_{\alpha}}{\sum_{\alpha=1}^A Mos_{\alpha}}$
- Then subsample elements at rate $\frac{b^*}{Mos_{\alpha}}$
- Why b^* ?
 - The actual number of second stage units selected is unknown; it's a target subsample size
 - If $Mos_{\alpha} = B_{\alpha}$, then $b^* = b$

Selecting units with PPeS

- Returning to employees example, suppose $b^*=18$, $a=2$, and
- $a = 2$ clusters are selected $\sum_{\alpha=1}^{10} Mos_{\alpha} = 2160$
- One subsample of expected size $b^* = 18$ each

$$f = \frac{2Mos_{\alpha}}{\sum Mos_{\alpha}} \times \frac{18}{Mos_{\alpha}} = \frac{36}{2160} = \frac{1}{60}$$

- With PPeS sampling, subsampling at a specified rate and not selecting a fixed number of elements from each selected cluster

Example

Unit	Last payroll	Now
1	443	460
2	162	172
3	127	130
4	554	554
5	115	125
6	291	310
7	64	68
8	70	74
9	232	246
10	102	141
Total	2160	2280

PPeS illustration

- Suppose $RN_1 = 702$ $RN_2 = 1744$

- Select Units 3 and 7

- For Unit 3,

$$\frac{b}{Mos_3} = \frac{18}{127} = \frac{1}{7.056}$$

- For Unit 7,

$$\frac{b}{Mos_7} = \frac{18}{64} = \frac{1}{3.556}$$

- If $Mos_\alpha = B_\alpha$, a sample of exactly size 36 will be selected

Sample size variation returns

- But here $Mos_{\alpha} \cong B_{\alpha}$
- For Unit 3, $B_{\alpha} = 130$, and $\frac{b}{Mos_3} = \frac{1}{7.056}$ yields

$$x_a = \frac{b}{Mos_{\alpha}} \times B_{\alpha} = \frac{1}{7.056} \times 130 = 18.425$$

- For interval 7.056, select a sample of **18** employees with probability 0.575
- Or a sample of **19** with probability 0.425

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B. Stratification

- Mos , a , b^* , etc., are specified within strata

$$f = f_h = \frac{a_h Mos_{h\alpha}}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}} \times \frac{b_h^*}{Mos_{h\alpha}} = \frac{a_h b_h^*}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}}$$

- Retain *epsem* for stratified PPeS sampling across strata: $f = f_h$ for all h .

Implicit stratification

- Systematic PPeS sampling implicitly stratifies by selecting within each zone one subsample size b^*
- Stratification notation not necessary with this design
- Zone size (or sampling interval) is

$$\frac{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}}{a_h}$$

Select $a = 4$ units

- $a = 4$ subsamples of size $b^* = 18$ from $Mos_t = 2160$

- Then

$$f = \frac{ab^*}{\sum Mos_{\alpha}} = \frac{4 \times 18}{2160} = \frac{1}{30}$$

- Zone size is $2160/4 = 540$

Explicit stratification

Stratum I		Stratum II	
Unit	Mos	Unit	Mos
1	443	5	115
2	162	6	291
3	127	7	64
4	554	8	70
		9	232
		10	102
Total	1286		874

Paired selection

$$f_1 = \frac{1}{30} \neq \frac{2Mos_{1\alpha}}{\sum_{\alpha=1}^{A_1} Mos_{1\alpha}} \frac{18}{Mos_{1\alpha}} = \frac{36}{1286}$$

$$f_2 = \frac{1}{30} \neq \frac{2Mos_{2\alpha}}{\sum_{\alpha=1}^{A_2} Mos_{2\alpha}} \frac{18}{Mos_{2\alpha}} = \frac{36}{874}$$

- Can manipulate $a_h, b_h^*, \sum_{\alpha=1}^{A_h} Mos_{h\alpha}$
- Uncommon to vary a_h

Vary subsample size

- For paired selections,

$$f_h = \frac{2Mos_{h\alpha}}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}} \frac{b_h^*}{Mos_{h\alpha}}$$

- Adjusting the subsample sizes across strata,

$$f_1 = \frac{2Mos_{1\alpha}}{\sum_{\alpha=1}^{A_1} Mos_{1\alpha}} \frac{b_1^*}{Mos_{1\alpha}} = \frac{1}{30}$$

$$b_1^* = 21.43$$

$$b_2^* = 14.57$$

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C. Oversize units

- Oversize units are clusters for which $Mos_{h\alpha}$ is so large that the unit has a certain chance of selection
 - Often so large that they will be selected multiple times in systematic PPeS
- For example, Unit 4 has $Mos_4 = 554 > 540$
 - Selected twice with probability

$$\frac{Mos_4}{540} = 1.0259$$

Solutions

- If there are few such clusters and chances of multiple selections small leave them in the list
 - If one selected k times, select k subsamples
- If there are many such clusters comprising a large share of the population, place in separate strata
 - Each such cluster is now a stratum, a cluster selected with certainty
 - Sampling rate(s) are applied directly within clusters
 - Self-representing units (SRU)

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D. Undersize units

- Undersize units are clusters for which $Mos_{h\alpha} < cb^*$
- $Mos_{h\alpha} < b^*$ implies sampling within cluster at rate > 1
- All elements must be selected in the cluster, and

$$f = \frac{a_h Mos_{h\alpha}}{\sum Mos_{h\alpha}} \times 1$$

Solutions

- Link undersize units to form linked units of minimum sufficient size
 - Link before selection for the entire frame, especially if the frame is a computerized list
 - Can create clusters with greater heterogeneity
 - In area sampling, link geographically contiguous units
- Linking after selection
- If numerous, place in separate stratum
- Undersize units can include zero measure clusters

Linking after (during) selection

- (1) Identify selected unit (block). If the selected unit and next on the list are minimum sufficient size, **STOP**
- (2) If selected unit or next are not of minimum sufficient size,
 - a) **Move forward** in list until first unit of minimum sufficient size is encountered
 - b) **Cumulate units backwards** until a linked unit of minimum sufficient size is created
 - c) Continue process until the selected unit is linked

Example

Tract	Block	HU's
28	210	23
	211	17
	212	36
	213	1
29	101	0
	102	0
	103	0
	104	0
	105	1
	106	0
	107	0
	108	0
	109	0
	110	0
	111	0
	112	0
	113	0
	114	60
	115	9

Example

Tract	Block	HU's
28	210	23
	211	17
	212	36
	213	1
29	101	0
	102	0
	103	0
	104	0
	105	1
	106	0
	107	0
	108	0
	109	0
	110	0
	111	0
	112	0
	113	0
	114	60
	115	9

Link 101-113, 212-213, size = 38

Link 105-113, size = 1